# Nina Knitowski

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## EDUCATION

**Columbia Mailman School of Public Health** May 2027
MPH in Epidemiology, Certificate: Biostatistics

**University of California Los Angeles (UCLA)** March 2025
BS Human Biology and Society GPA: 3.94/4.0, *Cum Laude Honors*

## SKILLS

**Data Analysis and Statistics**: R, Python, SAS
**Computer**: Microsoft Word, Excel, PowerPoint
**Language**: English, proficiency in Spanish

## EXPERIENCE

**Executive Administration Intern -** *NYC Health and Hospitals Elmhurst Hospital* June 2026-August 2026

- Support the Chief of Ancillary Services in overseeing operations for 12 clinical departments within a major metropolitan hospital system.
- Lead a departmental initiative in Radiology to integrate a new database system, effectively bridging training gaps and improving staff competency tracking.
- Spearhead the digitization and reorganization of sensitive HR personnel files, enhancing record accessibility and ensuring compliance with hospital data standards.

**Neuroscience Student Researcher -** *UCLA Jessica Rexach & Karen Gylys Lab* January 2022-June 2025

- Investigated and compared isolation methods for extracellular vesicles from organoid media to develop an effective protocol from size exclusive chromatography to immunoprecipation using magnetic beads that specifically bound to extracelular media
- Evaluated exosomal release in brain-derived organoids with and without point mutation R406W that is known to contribute to spread of tau pathology
- Analyzed genetic and regional differences in exosomal release of brain derived organoids
- Collaborated with other laboratories and balanced project goals with feasibility, limited resources, and varying areas of expertise
- Quantified and presented findings in lab meetings and broader audiences such as undergraduate research week

## VOLUNTEER AND ADDITIONAL PUBLIC HEALTH EXPERIENCE

**Read Ahead Volunteer -** *Read Ahead PS 67 School* September 2025-Present

- Partnered with 4th grader and give individualized help to promote literacy weekly through school year

**Athletic Mentor and Tutor**- *Community Public Health Literacy (CIPHER)* September 2025-Present

- Tutored elementary school students in STEM to build academic confidence and foundational literacy.
- Empowered young girls through athletic mentorship centered on the motto “Go Girls Go.”

**Emergency Medical Technician (EMT) -** *Millville Fire Department Delaware* June 2025-Present

- Responded to 911 emergency calls, providing rapid assessment, life-saving care, and safe transport for patients in high-stress situations in the Southern Delaware area with medical protocols and HIPAA standards.

**Helpline Volunteer -** *Alzheimer’s Los Angeles* March 2025-Present

- Providing compassionate support and resources to patients and families affected by Alzheimer’s Disease (Remote).

**Pediatric Literacy Volunteer -** *Venice Family Clinic* February 2025-June 2025

- Promoting early childhood literacy in low-income communities.

**Emergency Medical Technician (EMT) -** *PRN Ambulance Los Angeles, CA* April 2024-June 2025

- Skilled in patient assessments, intermediate medical care, and inter-facility transfers, with expertise in basic interventions, effective communication, and accurate documentation.

## LEADERSHIP

**Vice President-** *Columbia Mailman School of Public Health Journalism Club* Sept 2026-Present

- Podcast Host (2025–2026) and Executive Podcast Lead *“Kind of a Big Health Deal”*
- Founded a student-run podcast to advance public health literacy by addressing critical challenges, including healthcare affordability, food insecurity, care access, and neighborhood mortality disparities across NYC.
- Produced and hosted a debut mini-series featuring interviews with civil rights advocates Dr. Robert Fullilove and Dr. Heather Butts, Nobel Laureate Dr. Harold Varmus, global health leader Nina Schwalbe, and helped organize a student round table

**Campus Ambassador -** *Black Sheep Marketing* August 2022-March 2023

- Worked on campaigns with various companies who wished to expand their marketing endeavors by distributing advertisements in greater Los Angeles Area.

**Mentor -** *Society and Genetics Club* September 2024-March 2025

- *As* a senior, paired with 3 underclassmen in the same major. Tasked at guiding them through their academic pursuits throughout college through weekly meetings.

## PUBLICATIONS

Melnik, M., Mitri, V., Atluri, S., **Knitowski, N**., Kodavatikanti, A., Bilousova, T., Rexach, J. E., & Gylys, K. H. (2024)**.** Human brain microglia responses to amyloid beta and tau pathology in high pathology controls and across disease stages. *Alzheimer's & Dementia: The Journal of the Alzheimer's Association.* https://doi.org/10.1002/alz.088080

Flow cytometry detection of inflammatory markers in glial cells and synaptosomes derived from AD cortex (2023)
Melnik, M., Mitri, V., Atluri, S., **Knitowski, N.**, Kodavatikanti, A., Bilousova, T., & Gylys, K. H. (2023). Phenotypic status explored in parietal cortex of Alzheimer's patients and non-demented controls that vary on the spectrum of neuropathological AD. *Alzheimer's & Dementia: The Journal of the Alzheimer's Association.* https://doi.org/10.1002/alz.073956

Bilousova, T., Melnik, M., Matsuo, Y., Kodavatikanti, A., **Knitowski, N**., Atluri, S., & Gylys, K. H. (2023). Brain extracellular vesicles from Alzheimer's disease and Down syndrome patients mediate spread of senescence-like phenotype in vitro. *Alzheimer's & Dementia: The Journal of the Alzheimer's Association*. https://doi.org/10.1002/alz.077514

## EXTRACURRICULAR ACTIVITIES

Mailman: Student Enviormental Association member (2025-2027), Columbia Mail School of Public Health Journalism Club member ((2025-2027) and Vice President (2026-2027), CIPHER club member (2025-2027)
UCLA: Society and Genetics Club Member and Mentor (2023-2024), Daily Bruin: Podcast Intern (2022), Los Angeles Club

## CERTIFICATIONS

Completion of EMT course (180 hours)
Completed NREMT and PA psychomotor exam
Certified in the state of Delaware and California
Basic Life Support/ CPR Provider November 2025
Delaware EVO Driver's License
Certification in Thorndale, PA December 2022

## ACHIEVEMENTS

**Mailman School of Public Health Wyman Scholar** 2025-2027
Selected for an academic scholarship for high scholastic ability for public health careers (academic, research, service)

**UCLA Society and Genetics Case Competition Winner** Spring 2024
Across Human Biology and Society Undergraduates, presented about Ozempic and the role of consumerism in health and wellness through bioethical analysis.

**Undergraduate Research Week** May 2023, 2024
Invited to present in front of professors and peers on my project titled "Investigating the Role of Exosomes in the Spread of Tau Pathology in Neurodegenerative Disorders such as Alzheimer's."

**Phoenix Award** July 2025
Award by Sussex EMS in recognition of saving a patient's life by completing CPR when a patient is confirmed to be in cardiac arrest.